

# Online Assessment and Control of Transient Oscillations Damping

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**Abstract**—A new approach to online assessment and control of transient oscillations is proposed. It relies on coupling Prony analysis and the SIME transient stability method, with the twofold objective: on one hand, to improve Prony's applicability and reliability, and on the other hand, to design generation rescheduling patterns able to enhance the damping of poorly damped oscillations and/or stabilize them. Simulations performed on two different systems illustrate the technique and show main features: accuracy, ability to uncover and assess the influence of system machines' generation on damping, and compatibility with real-time requirements. Comparison of the results obtained by the proposed approach with conventional eigenanalysis techniques is provided. Additional method's interesting byproducts are also discussed.

**Index Terms**—Control of poorly damped oscillations, damping assessment, modal identification, Prony analysis, SIME method, transient stability.

## I. INTRODUCTION

THE operation of various power systems has been limited by system oscillations [1]. In systems experiencing undamped oscillations problems, *damping performance criteria* are set up to classify the oscillations as "secure" or "insecure." These criteria are expressed in terms of damping ratio or of the time needed for an oscillation to be damped to a percentage (say 50%) of the first swing oscillation. (Examples of damping criteria can be found in [1].)

This complex problem has usually been tackled by offline studies performed in the planning and operational planning contexts. To enhance power system damping, planning studies usually add and tune damping controllers, like supplementary signals from power system stabilizers to the excitation control of generators, the voltage control of flexible ac transmission systems (FACTS) devices like static var compensators, or the modulating controls of high voltage direct current lines. However, even with appropriate controllers in place, the operator can face system conditions which fall beyond what the controllers are designed for [2], especially in the context of electric industries restructuring, given the increasing complexity of their dynamics resulting from the interconnection of power utilities and their

operation near to their limits. Therefore, additional online remedial measures are necessary to ensure system security.

Online assessment and control of transient oscillations is thus becoming an issue of great concern in the operation of power systems. Today, the need is felt to include these techniques as part of a transient stability function, which, besides exploring whether the system is able to withstand the occurrence of a contingency, should also ascertain that the transient oscillations caused by such a contingency meet preassigned damping performance criteria and have little impact on the quality of service [3], [4].

An online technique for assessing system damping has recently been proposed, using single or multichannel Prony algorithms, which consider one or several generator swing curves at a time [5], [6]. However, a main difficulty of this approach is that the accuracy of the results strongly depends on the choice of the generator curves, and that Prony's multichannel program is limited to the simultaneous analysis of a small number of curves only (e.g., 4 in [20]). Since power systems are usually composed of hundreds of generators, this makes the choice of generator curves problematic. To circumvent this difficulty, [7], [8] proposed to couple Prony analysis with the single machine equivalent (SIME) transient stability method [8]–[12].

As explained in Section II, SIME is able to compress the dynamic behavior of the multimachine system into that of a one-machine infinite bus system, making the application of single channel Prony analysis to large real systems possible.

The preliminary study [7] has shown that the coupling of SIME with a Prony algorithm is indeed able to characterize the damping and nature of power system oscillations excited by a contingency, by precisely determining the dominant modes of oscillation and the system machines that have a major influence on the ensuing oscillatory phenomena.

This paper revisits this preliminary study and complements it with a control technique able to improve transient oscillations damping by proper rescheduling of precontingency power system generation [8]. The resulting integrated damping assessment and control technique is explored on two different power systems, exhibiting two different modes of oscillation: a plant mode on the EPRI 627-machine system, and an interarea mode on the EPRI 88-machine system. Conventional eigenanalysis studies validate these results.

## II. SIME'S UNDERPINNINGS

NB. This section borrows material from [8], [9], in order to introduce notation and to make the paper self-reliant. The illustrations refer to simulations performed in Section VI.

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### A. Foundations

SIME conjectures that the synchronism of a multimachine power system is lost as soon as its machines are irrevocably split into two groups [9]<sup>1</sup> and, further, that its dynamics may be inferred from the dynamics of the one-machine infinite bus (OMIB) equivalent to these groups. The OMIB parameters are computed from the multimachine power system parameters, furnished by a time-domain program, and refreshed at each step of this program. The dynamics of this “generalized time-varying” OMIB [13] are explored until reaching the instability conditions (1) provided by the equal-area criterion (see below).

In short, the continuous interplay between a time-domain program and the equal-area criterion allows identifying the groups of critical and noncritical machines and the corresponding OMIB. The procedure main steps are briefly described below and illustrated in Fig. 1(a) and (b) drawn for the 627-machine system simulated in Section VI.

Observe that since SIME uses the output data of a transient stability program, it allows for the detailed representation of the power system, just like the program with which it is coupled [8]–[10].

### B. Identification of the Critical Machines, Instability Conditions and OMIB Parameters

To identify the system critical machines (CMs) relative to a stability scenario, SIME starts exploring the output data of a time-domain transient stability simulation as soon as the system enters its post-fault phase, considering a few candidate decomposition patterns and resulting candidate OMIBs. The parameters of these candidates  $\delta$ ,  $\omega$ ,  $M$ ,  $P_m$ ,  $P_e$ , (for angle, speed, inertia coefficient, mechanical power and electrical power, respectively) are computed from corresponding individual machines parameters. For example, in Fig. 1(b), the OMIB trajectory is plotted from the multimachine trajectories (swing curves) of Fig. 1(a).

The candidate OMIB which first reaches the early termination instability conditions of the equal-area criterion

$$P_a = 0; \quad \dot{P}_a > 0 \quad (1)$$

is declared to be the critical OMIB of concern or simply the OMIB [8], [9].

In the above expression,  $P_a$  denotes the accelerating power, difference between OMIB’s mechanical and electrical powers

$$P_a = P_m - P_e. \quad (2)$$

In what follows, use will also be made of the OMIB angle  $\delta_u$  and speed  $\omega_u$  assessed at the “time to instability”  $t_u$ ; this is the time where the above instability conditions are met. At this time, SIME stops the time-domain simulation, and declares the critical machines to be those of the “advanced group of machines” of the critical OMIB. The stability margin is then calculated at this time by the EAC as

$$\eta_u = -\frac{1}{2}M\omega_u^2. \quad (3)$$

<sup>1</sup>The machines of each group are not supposed to be coherent.

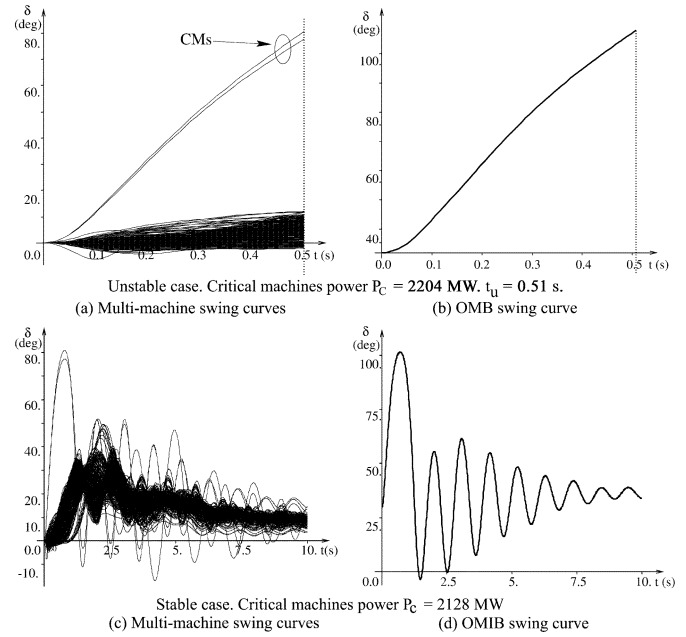


Fig. 1. Unstable and stable cases simulated on the EPRI 627-machine system. Clearing time of both cases: 67 ms.

### C. Technicalities and Functionalities

- 1) Strictly speaking, OMIB and CMs are determined on unstable scenarios only. However, by continuation, they are supposed to be still valid on a stable scenario too, provided that the stability conditions of the two scenarios are “close enough” to each other (see 1 of Section IV-B-II). For example, Fig. 1(d) displays the OMIB swing curve of the stabilized multimachine case of Fig. 1(c); the OMIB structure was determined on the unstable case of Fig. 1(a), while its parameters are computed from those of the stabilized multimachine simulation.
- 2) Besides the instability conditions (1), the equal-area criterion provides also early termination stability conditions, which, however, are less useful. Indeed, to avoid missing multiswing instabilities, the simulation must be carried out on the maximum integration period.
- 3) The interplay between time-domain simulation of the multimachine system and equal-area criterion applied to the resulting “generalized” OMIB [13] provides valuable assets to SIME-based methods: full flexibility with respect to power system and contingency modeling, ability to identify the critical machines, etc. More generally, the compressed information of the OMIB dynamics allows elaborating new techniques, like control of flexible ac transmission system (FACTS) devices using a control Lyapunov function with control input signal provided by SIME [14], or the damping assessment and control proposed in this paper.

### III. DAMPING ASSESSMENT BY PRONY ANALYSIS

#### A. Foundations

Prony's method fits a series of damped sinusoids, to an observed function  $y(t)$  [1], [15]

$$\hat{y}(t) = \sum_{i=1}^q A_i e^{\alpha_i t} \cos(2\pi f_i t + \phi_i) = \sum_{i=1}^N B_i e^{\lambda_i t} \quad (4)$$

where, for each one of the  $q$  components of the estimated signal  $\hat{y}(t)$ ,  $A$  is the amplitude,  $B$  is the signal residue,  $\alpha$  is the damping,  $\lambda = \alpha \pm j\omega$  is the associated eigenvalue,  $f$  is the frequency, and  $\phi$  is the phase angle. The damping ratio defined by

$$\zeta = \frac{-\alpha}{\sqrt{\alpha^2 + \omega^2}} \quad (5)$$

is used for measuring the rate of decay of the amplitude of the oscillation.

The approximating function has a structure very similar to that of Fourier series, but the basis functions are more general. They can be damped and their frequencies need not be related harmonically or to the duration of the signal. Further, sample points of the signal should be equally spaced.

The reconstructed signal does not usually fit the observed one exactly. An appropriate measure of the quality of this fit is the signal to noise ratio (SNR) defined as

$$\text{SNR} = 20 \log \frac{\|y(k) - \hat{y}(k)\|}{\|y(k)\|} \quad (6)$$

where  $\|\bullet\|$  denotes the root-mean-square norm and the SNR is expressed in decibels (dB).

#### B. Practical Considerations

In general, Prony analysis can be used to identify dominant system modes when applied to output data from transient stability programs, measurements from system tests or disturbances, or directly to ambient data [15].

[3], [16], [17] summarize some of the techniques currently used for the study and identification of electromechanical modes using Prony-based methodologies. When Prony analysis is applied to transient stability program simulations, they have to be close to linear, if a direct comparison with modal analysis is required. In a system record (or a signal obtained from transient stability simulation, like in the present work) this is often the case at the end of the data window, rather than at its inception, where the amplitude of the transients may be large [1].

Since Prony results may change with the record length and position, care should be taken to obtain accurate estimates to system behavior. In this process, additional modes can be encountered, needed to fit signal offsets and noise [15].<sup>2</sup> A sampling technique, known as "sliding window solutions" is useful for detecting changing signal characteristics (due to nonlinearities or hidden inputs) and for distinguishing essential modes

<sup>2</sup>Signal offsets generally produce modes near 0 Hz or near the Nyquist frequency,  $1/(2\Delta t)$ , where  $\Delta t$  is the signal sampling rate [15].

from those that are mere accessories to the fitting process [16]. For more details, please refer to [17].

#### C. Alleviating Current Methodologies Weaknesses by SIME

Severe limitations are nonetheless inherent to the procedure aforementioned. First, the accuracy of the results is dependent upon the choice of the generator curves. Since power systems are usually composed of hundreds of generators this choice may become problematic. Also, Prony results obtained for each individual generator can be affected by nonlinearity effects, which in turn, results in conflicting modal estimates at different system locations [5], [6]. In addition, the selected machines may participate in several local or inter-area modes; the corresponding analysis of specific system modes of concern is usually costly in terms of computer time. Other aspects such as the effects of time varying dynamics, hidden inputs and noise components may further complicate Prony analysis especially when processing ambient data [5].

Solutions have been proposed to alleviate the first and second problems, particularly the use of multiple signals [5], [6]. More recently, the authors have addressed these aspects by coupling Prony analysis with the SIME method [7], [8]. This approach enables the relevant dynamics of the system following a system perturbation to be singled out and helps in alleviating some of the above problems.

In particular, by concentrating on the relevant dynamic of the system provided by SIME, Prony accuracy and robustness can be improved. As explained above, SIME produces an instantaneous second-order dynamic equivalent of the system in which the key dynamics of machines exchanging energy is contained. Unlike other approaches, the OMIB rotor angle inherently captures the nonlinear and nonstationary dynamics of the system and can be used to determine linear and nonlinear attributes of system oscillations. Also, because of its formulation, the OMIB signals are less likely to include other less relevant dynamics, which greatly facilitates the analysis of the slow dynamics of interest associated with the critical inter-area modes.

The approach elaborated in Section IV enables the key underlying dynamics of the system to be identified, reduces computing cost for the study of large complex systems, and provides damping control techniques.

### IV. PROPOSED APPROACH

#### A. Generalities

As aforementioned, to assess damping, Prony analysis is applied to the OMIB signal provided by SIME. On the other hand, to stabilize unstable oscillations—or improve poorly damped ones, generation is shifted from critical to noncritical machines, according to the SIME-based transient stability control technique [8], [11].

In principle, the oscillations damping computation function is part of an integrated software, which in a sequence:

- a) identifies the harmful contingencies (i.e., the unstable scenarios), out of a list of postulated contingencies, using the contingency filtering, ranking and assessment (FILTRA) software [10];

- b) stabilizes them using the transient stability control software [11];
- c) assesses the damping ratios of the stabilized scenarios to decide whether they meet the damping criteria; if not, control is further pursued.

The integrated software is displayed in Fig. 9 and discussed in Section VIII. In this section, we simplify descriptions by considering an isolated contingency, for which a full transient stability assessment is conducted, followed by a damping assessment and damping control cycle.

## B. Damping Assessment

1) *Procedure:* For a given contingency:

- i) run SIME to compute the transient stability limit (critical clearing time or power limit), using one stable simulation in addition to unstable ones;
- ii) on the least unstable simulation, identify the critical machines (CMs) and corresponding OMIB as soon as the time to instability  $t_u$  is reached;
- iii) on the stable simulation used in i), perform Prony analysis to assess the damping ratio  $\zeta_0$  of the OMIB swing curve. If this ratio is not good enough, store the total generation power of the critical machines  $P_{C0}$ ;
- iv) at time  $t_u$  of the least unstable simulation, determine also the “relative relevance” of each one of the CMs, defined as the product of its inertia times its angle (absolute or relative to a reference) [9], [11]. This is a useful measure of the impact (influence) of each CM on the power system damping and can be used to improve it, if necessary.

2) *Applicability Conditions and Specifics:*

- 1) For a given contingency, a pair of stable and unstable scenarios “close enough to each other” may be obtained by conducting two simulations: one just above and one just below the transient stability limit (critical clearing time or power limit).

The computation of such a limit relies on pair-wise extrapolation (or interpolation) of stability margins. Note that this iterative procedure generally reduces to 1–2 unstable and 1 stable simulations, thanks to the quasilinear variation of margins with stability conditions [8], [9], and to the criteria proposed in [9, p. 98] for choosing proper initial stability conditions. Observe that, anyhow, unstable simulations are little time consuming, since they require a time-domain simulation only up to the time to instability  $t_u$  which, generally, is much shorter than the maximum integration period (MIP).

- 2) For a list of contingencies, where the damping assessment and control tasks become parts of the general organization of Fig. 9, the above procedure is further simplified.
- 3) The method is designed to assess and control damping of nearly unstable contingency scenarios. Given that these are the most interesting ones, the above condition of item Nr 1 is not very restrictive.
- 4) The equal-area criterion is not directly involved in the damping assessment procedure, but plays a key role, as in all SIME-based methods: it is the core of the identification of the system mode of separation (i.e., of the critical and

noncritical machines and, hence, of the OMIB structure); besides, it detects the time to instability  $t_u$  when the multimachine system loses synchronism and SIME stops the simulation.

- 5) The shape of the OMIB swing or  $P$ - $\delta$  curve may differ significantly from the shape of an actual machine. Indeed, the OMIB does not describe an actual machine’s behavior, but, rather, compressed information about the multimachine system dynamics.

## C. Damping Control

1) *Procedure:* The principle is the same for stabilizing otherwise unstable oscillations or for improving poorly damped, though stable ones.

- i) For the stable simulation used in iv) of Section IV-B-I, consider a small amount of generation decrease in critical machines identified in ii) of Section IV-B-I and let  $\Delta P_C$  be this decrease.
- ii) Increase by the same amount the generation in noncritical machines (NMs).
- iii) With this new generation pattern<sup>3</sup> run a power-flow program to compute the new prefault operating conditions, then SIME to determine the structure and parameters of the new OMIB.
- iv) Use Prony analysis to assess the damping ratio of this new OMIB,  $\zeta_I$ : if it complies with prespecified damping criteria, stop; otherwise, go to i) and apply an additional generation shift.

2) *Specifics:*

- 1) Admittedly, shifting generation from one group of machines to the other may cause a change in CMs, especially if the electrical distance between the two groups is small, as in some interarea oscillations.

However, generally, there will be a subset of machines that are consistently critical (possibly the most advanced ones). In such cases, it might be convenient to shift generation from some critical machines only.

- 2) Changes in CMs affect the number of iterations and, hence, computing time. This is especially true for stable simulations (which are performed on the entire integration period to avoid missing multiswing instabilities), whereas unstable simulations are stopped at  $t_u$  and are little time consuming.

The simulations of Section VII and ensuing discussions of Section VIII illustrate these considerations.

## V. SIMULATION CONDITIONS

### A. General Description

Simulations have been performed on two real-world power systems: the EPRI test system A, comprising 4112 buses, 6091 lines, and 627 machines, and the EPRI test system C comprising 434 buses, 2357 lines, and 88 machines [18].

The power systems are modeled in their usual, detailed way, representing the synchronous machines in the areas of interest by their full Park equations, and controls. The EPRI

<sup>3</sup>The generation rescheduling of CMs and NMs can comply with various additional technical or economical requirements (e.g., see [11]).

627-machine system includes, in addition, dynamic and static load models for some important load buses [18].

The contingencies considered are three-phase short-circuits applied at a bus and cleared by tripping one or several lines.<sup>4</sup> They yielded oscillatory plant-modes (627-machine system) and interarea modes (88-machine system). In all simulations, the SIME method was coupled with the ETMSP program [19].

To assess damping, two programs were used: the spectrum function of the EPRI output analysis program [19] for plant mode oscillations, and the dynamic system identification toolbox [20] for interarea mode oscillations. Damping analysis relies on simulations obtained according to the procedure of Section IV-B, performed over a 10-s period. In addition, conventional eigenvalue analyses were carried out for an extensive load range to establish the limits of validity of the proposed approach.

### B. Objectives

The objectives pursued by the simulations were manifold:

- to validate the modal identification procedure on realistic situations;
- to illustrate the application of this procedure on the two types of oscillations;
- to get more insight into similarities and specifics of the oscillatory phenomena;
- to validate the damping control procedure;
- to appraise the performance of the techniques;
- to provide insight into the kinds of dynamic phenomena that may arrive in transient oscillations.

## VI. PLANT-MODE OSCILLATIONS

### A. Damping Assessment

The 627-machine system was subject to a contingency that creates plant-mode oscillations, involving two critical machines (see Fig. 1).

Prony analysis was applied to the OMIB curve of Fig. 1(d), whose structure was defined on the unstable simulation displayed in Fig. 1(a) and (b).

An increasing number of sample points and decreasing sampling rate were used until reaching a satisfactory signal to noise ratio (SNR of about 40 dB [17]). Three such sets of simulations have been performed for various sliding windows. Table I summarizes the resulting damping features obtained with each one of these sets, found for the best window.

As shown in this table, the computations performed with 250 data points and sampling ratio of 10 ms provide the best results (largest SNR). Note that, anyhow, the damping features provided by the three sets of simulations are quite close to each other.

As a Prony validation test, Fig. 2 compares the original OMIB curve with the curve estimated by Prony for the window 4.7–7.1 s, presented in Table I. The two curves fit almost perfectly.

<sup>4</sup>More precisely, contingency # 8 was applied to the 627-machine system and contingency # 22 to the 88-machine system [18].

TABLE I  
OMIB'S PRONY ANALYSIS FOR SEVERAL DATA POINTS AND SAMPLING RATES (SRs). 627-MACHINE SYSTEM; CONDITIONS OF FIGS. 1(D) AND 2 [7],[8]

| Period (s)                     | SR (ms)   | $\alpha$ (rad/s) | $\omega$ (rad/s) | f (Hz)       | $\zeta$ (%) | SNR (dB)     |
|--------------------------------|-----------|------------------|------------------|--------------|-------------|--------------|
| Nr of data points: approx. 125 |           |                  |                  |              |             |              |
| 4.0–9.0                        | 40        | -0.423           | 5.88             | 0.936        | 7.16        | 34.28        |
| Nr of data points: approx. 250 |           |                  |                  |              |             |              |
| 4.0–8.9                        | 20        | -0.418           | 5.88             | 0.937        | 7.09        | 34.17        |
| Nr of data points: approx. 250 |           |                  |                  |              |             |              |
| <b>4.7–7.1</b>                 | <b>10</b> | <b>-0.483</b>    | <b>5.84</b>      | <b>0.930</b> | <b>8.23</b> | <b>42.35</b> |

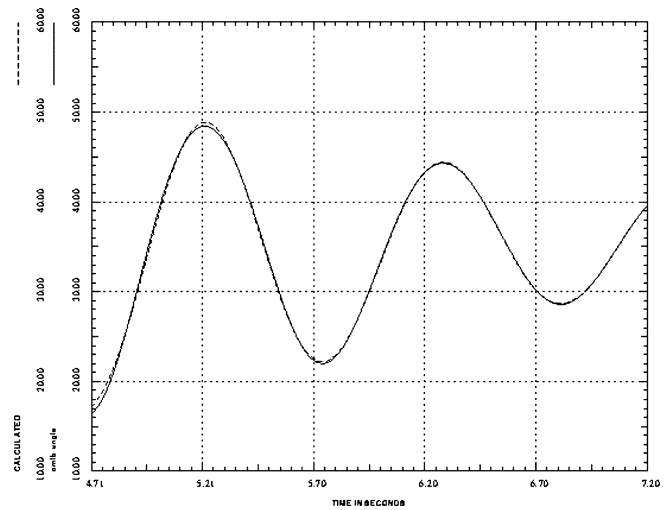


Fig. 2. Original OMIB and Prony curves estimated for the stable case of Figs. 1(d) and 3. [7], [8].

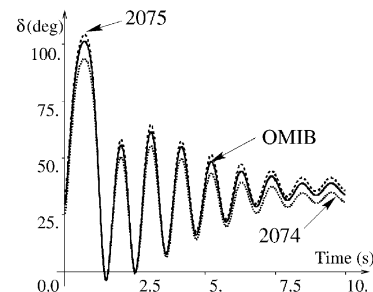


Fig. 3. Swing curves of OMIB and machines 2075 and 2074. EPRI 627-machine system (stable case) [7], [8].

As another validation check, Prony analysis was performed on the two relevant machines identified by SIME, (labeled 2075 and 2074), using the best conditions determined above. Fig. 3 portrays the OMIB and critical machines' swing curves, while Table II gathers main features. Observe that the frequency of the three curves is almost the same, while the value of the damping ratio of the OMIB is in between that of machines 2075 and 2074. These numerical results reflect well the shapes and relative positions of the three curves.

*Remark:* The two critical machines identified on the unstable simulation of Fig. 1(a) are also easily identified on the stable simulation of Fig. 1(c): hence, the information provided by SIME is not indispensable to Prony analysis; rather, it illustrates and validates the proposed approach.

TABLE II  
PRONY ANALYSIS OF OMIB AND MACHINES 2075 AND 2074. EPRI  
627-MACHINE SYSTEM (STABLE CASE). NUMBER OF DATA POINTS: 250;  
SAMPLING RATE: 10 MS; PERIOD OF STUDY: 4.7–7.1 s [7], [8]

| $\alpha$ (rad/s) | $\omega$ (rad/s) | $f$ (Hz) | $\zeta$ (%) | SNR (dB) |
|------------------|------------------|----------|-------------|----------|
| OMIB             |                  |          |             |          |
| -0.483           | 5.84             | 0.930    | 8.23        | 42.35    |
| Machine # 2075   |                  |          |             |          |
| -0.509           | 5.83             | 0.927    | 8.7         | 43.27    |
| Machine # 2074   |                  |          |             |          |
| -0.474           | 5.83             | 0.928    | 8.0         | 35.89    |

### B. Damping Control

As stated in Section IV-C, similar procedures are used to control poorly damped or undamped oscillations and transiently unstable cases [8]. More specifically, after redispatching machines' generation, damping is assessed by a Prony analysis of the OMIB signal; if the damping ratio is still too poor, additional rescheduling is applied. The iterative procedure is stopped when reaching the damping ratio desired.

As an example, consider again the case of Fig. 1, where two critical machines were identified. Decreasing the power of these machines ( $P_c$ ) from 2204 MW to 2128 MW stabilizes the system. The resulting damping ratio is found to be 8.23% (see Table I). Further, if the damping performance criterion for this system requires, for example, a damping ratio of 10%, additional power shift from critical machines must be performed. Table III shows a summary of the iterative procedure: power is decreased in steps of 100 MW and damping is assessed on the OMIB swing curve for each simulation. More precisely, column 1 displays the power of the critical machines, while columns 2 and 3 gather respectively the frequency (in Hertz) and damping ratio (in percent) of the oscillation. Finally, column 4 lists the signal-to-noise ratio in decibels for each simulation, in order to show the good agreement between the estimated and the original OMIB equivalent swing curves. The table shows that the system can thus reach the desired damping performance criterion after two iterations. On the other hand, Fig. 4 displays graphical results of the above simulations.

## VII. INTERAREA-MODE OSCILLATIONS

### A. Damping Assessment

Applying contingency # 22 to the EPRI 88-machine system creates an interarea-mode transient instability, stabilized by shifting generation from critical to noncritical machines. The swing curves of the unstable and stabilized cases are displayed in Fig. 5, while Table IV describes the stabilization process. In particular, column 3 shows that there are 36 critical and 52 noncritical machines, while column 5 indicates that to stabilize the system, 535 MW have been shifted; further, column 6 specifies that the system loses synchronism at 1.48 s, where the unstable simulation is stopped, while the stable simulation is explored on the maximum integration period (MIP) of 10 s.<sup>5</sup>

Prony analysis performed on the above stable case under optimal conditions (number of data points, sampling rate, SNR, and period of study) provides the results listed in Table V. This

<sup>5</sup>In column 6 of Table IV, MIP stands for "maximum integration period".

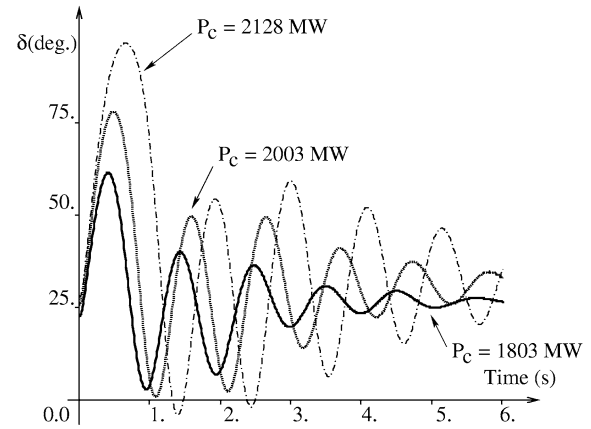


Fig. 4. OMIB swing curves for selected simulations of Table III [8].

TABLE III  
ITERATIVE DAMPING ENHANCEMENT OF STABLE CASE OF FIG. 1(D) [8]

| $P_c$ (MW)  | $f$ (Hz)    | $\zeta$ (%)  | SNR (dB)     |
|-------------|-------------|--------------|--------------|
| 2128        | 0.93        | 8.23         | 42.35        |
| 2003        | 0.93        | 9.9          | 57.00        |
| <b>1903</b> | <b>0.98</b> | <b>10.03</b> | <b>53.65</b> |
| 1803        | 0.98        | 11.51        | 40.21        |

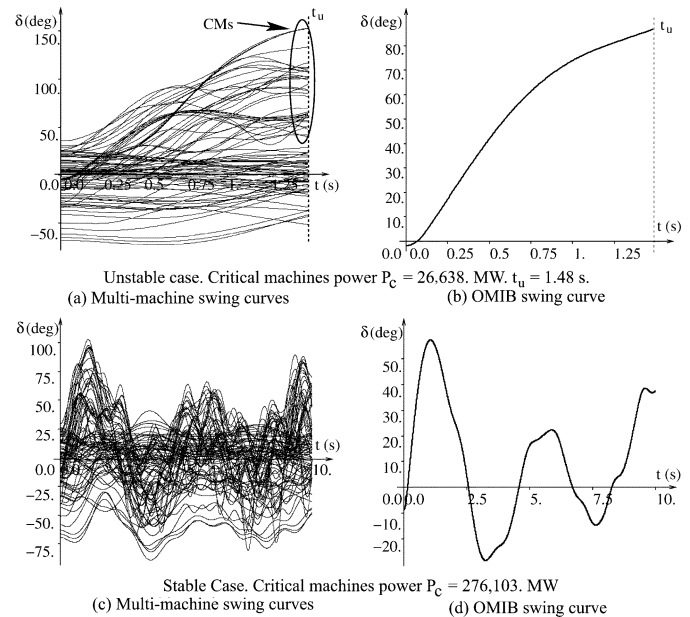


Fig. 5. Unstable and stable cases simulated on the 88-machine system. Clearing time of both cases: 120 ms.

TABLE IV  
STABILIZATION OF THE 88-MACHINE SYSTEM. MIP: 10 s

| It. # | $\eta$ (rad/s) <sup>2</sup> | Nr of CMs | $P_{ck}$ (MW) | $\Delta P_{ck+1}$ (MW) | $t_u$ (MIP) (s) |
|-------|-----------------------------|-----------|---------------|------------------------|-----------------|
| 1     | -0.08                       | 36        | 26 638        | -535                   | 1.48            |
| 2     | 0.69                        | ---       | 26 103        | ---                    | (10.00)         |

table shows that there are two dominant modes, and reveals that, actually, the above presumably "stabilized" case has negative damping ratio for one of them which, in turn, suggests that the corresponding oscillation is unstable and predicts system loss of synchronism.

TABLE V  
OMIB PRONY ANALYSIS FOR THE STABILIZED SCENARIO OF FIG. 5(D)

| Period<br>(s) | SR<br>(ms) | $\alpha$<br>(rad/s) | $\omega$<br>(rad/s) | f<br>(Hz)    | $\zeta$<br>(%) | SNR<br>(dB) |
|---------------|------------|---------------------|---------------------|--------------|----------------|-------------|
| 4.18-9.78     | 20         | -0.101<br>0.206     | 1.5117<br>5.5232    | 0.24<br>0.87 | 6.6<br>-3.7    | 42.29       |

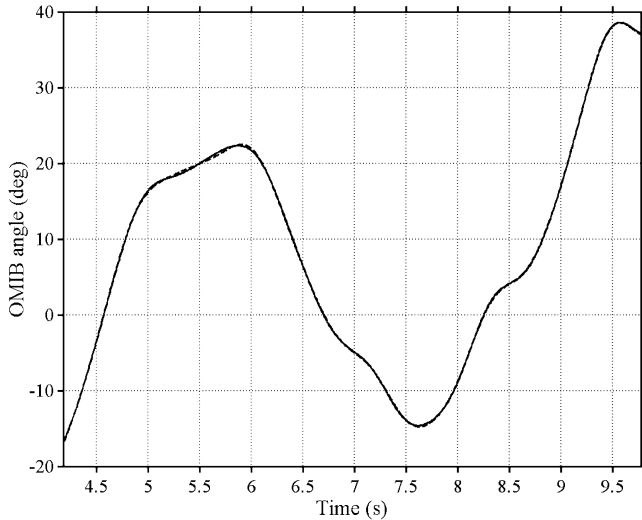


Fig. 6. Original OMIB and Prony curves for the stable case of Figs. 5(d) and 7. Number of data points: 281.

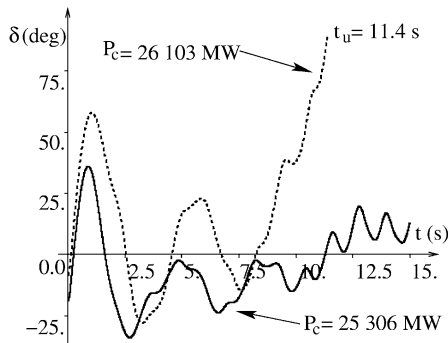


Fig. 7. Stabilizing an unstable interarea oscillation by decreasing the critical machines power from 26 103 MW to 25 306 MW. EPRI 88-machine system. Contingency clearing time of both cases: 120 ms.

Fig. 6, drawn for the “optimal window” ranging from 4.18 to 9.78 s shows the accuracy of the Prony damping assessment and suggests the imminence of instability. Actually, Fig. 7, drawn for a simulation conducted under the same conditions with Fig. 5(c) and (d) but for a maximum integration period of 15 s instead of 10 s, shows that the system loses synchronism at 11.4 s. A finer exploration, carried out in the context of damping control, allows shedding more light into these seemingly peculiarities. This is described below.

### B. Damping Control

To improve the damping of the above “stabilized” case until getting positive damping ratios for both dominant modes, a damping control procedure is carried out by shifting, as usual, generation from critical to noncritical machines.

TABLE VI  
DAMPING CONTROL ON THE 88-MACHINE SYSTEM. MIP: 10 s

| $P_C$ (MW)    | f (Hz)                     | $\zeta$ (%)              | SNR (dB)     |
|---------------|----------------------------|--------------------------|--------------|
| 26 103        | 0.24<br>0.87               | 6.6<br>-3.7              | 42.29        |
| 25 838        | 0.26<br>0.87               | 8.0<br>-3.5              | 41.63        |
| 25 572        | 0.27<br>0.87               | 4.7<br>-2.8              | 42.34        |
| <b>25 306</b> | <b>0.29</b><br><b>0.89</b> | <b>8.5</b><br><b>0.9</b> | <b>41.05</b> |

TABLE VII  
TRANSIENT STABILITY CHARACTERISTICS FOR THE CASES OF TABLES IV AND VI ON THE 88-MACHINE SYSTEM MIP = 15 s

| $P_C$ (MW)    | $\eta$ (rad/s) <sup>2</sup> | Nr of CMs | $t_u$ (MIP) (s) |
|---------------|-----------------------------|-----------|-----------------|
| 26 638        | -0.08                       | 36        | 1.48            |
| 26 103        | -0.01                       | 37        | 11.4            |
| 25 838        | /                           | 40        | 13.5            |
| 25 572        | -0.001                      | 26        | 14.9            |
| <b>25 306</b> | <b>1.50</b>                 | ---       | <b>(15)</b>     |

Table VI synthesizes Prony analysis results of the cases considered in the iterative damping control process where the power is successively shifted from critical to noncritical machines in steps of approximately 1%. It shows that to get positive damping ratios requires decreasing the critical machines’ generation to 25 306 MW.

On the other hand, Table VII reports on transient stability simulation results for the above considered five cases (the one of Table IV, row 2 and the four of Table VI). Observe that extending the maximum integration period to 15 s makes the stability criterion (positive margin, see column 2 of Table VII) coincide with the damping one (positive damping ratios; see col. 3 of Table VI).

Fig. 7 displays the OMIB swing curves of the two extreme cases of Table VI: the “fully unstable” (having the largest negative damping ratio) and the “fully stabilized” ones, drawn for a maximum integration period of 15 s.

### C. Eigenvalue Analysis

Detailed eigenvalue studies were conducted to validate the correctness of the proposed method as well as to gain further insight into the nature of the dynamic phenomena described by the OMIB model. Table VIII summarizes the main characteristics of the slowest electromechanical modes, along with the associated frequency and damping computed at the postfault network condition for the 26 103-MW generation level. The analysis concentrates on the 0.26-Hz interarea mode 1 and the 0.89-Hz unstable interarea mode 14, which are associated with the transient behavior predicted by the OMIB model.

The nature of the dynamic behavior of these modes becomes evident from the analysis of the speed components of the corresponding right eigenvectors in Fig. 8.

Examination of the results presented in Fig. 8(a) illustrates that for interarea mode 1, machines in the critical cluster of most-advances machines swing in opposition to machines in the noncritical group as suggested by the analysis of the OMIB model (refer to Table IX). In contrast, the analysis of interarea mode 14 in Fig. 8(b) represents a rather localized phenomenon involving the interaction of several machines located in

TABLE VIII  
SLOWEST MODES OF THE EPRI 88-MACHINE SYSTEM

| Mode | $\alpha$ (rad/s) | $\omega$ (rad/s) | f (Hz) | $\zeta$ (%) |
|------|------------------|------------------|--------|-------------|
| 1    | -0.2266          | 1.639            | 0.261  | 13.69       |
| 2    | -0.2856          | 2.568            | 0.408  | 11.05       |
| 3    | -0.4973          | 2.736            | 0.435  | 17.88       |
| 4    | -0.4883          | 3.271            | 0.520  | 14.76       |
| 5    | -0.4987          | 3.528            | 0.561  | 14.00       |
| 6    | -0.4994          | 3.696            | 0.588  | 13.39       |
| 7    | -0.4961          | 3.931            | 0.625  | 12.52       |
| 8    | -0.4987          | 4.398            | 0.700  | 11.27       |
| 9    | -0.4921          | 4.572            | 0.727  | 10.70       |
| 10   | -0.4974          | 4.730            | 0.752  | 10.46       |
| 11   | -0.4799          | 5.106            | 0.812  | 9.36        |
| 12   | -0.3339          | 5.151            | 0.819  | 6.47        |
| 13   | -0.4998          | 5.412            | 0.861  | 9.19        |
| 14   | 0.0354           | 5.617            | 0.894  | -0.63       |

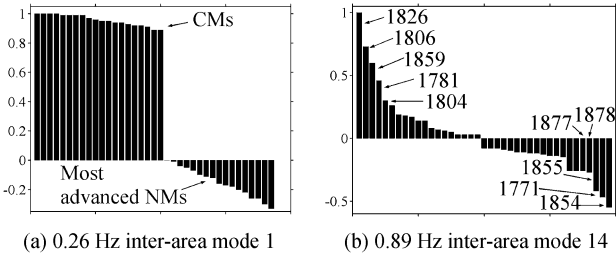

 Fig. 8. Mode shape of the dominant modes.  $P_C = 26103$  MW.

 TABLE IX  
RANKING (R) & IDENTIFICATION OF THE FIRST 20 CRITICAL MACHINES.  
EPRI 88-MACHINE SYSTEM;  $P_C = 26,103$  MW

| (R) Machine | (R) Machine | (R) Machine | (R) Machine | (R) Machine |
|-------------|-------------|-------------|-------------|-------------|
| (1) 1826    | (5) 1804    | (9) 1834    | (13) 1806   | (17) 1782   |
| (2) 1783    | (6) 3015    | (10) 1871   | (14) 1877   | (18) 2658   |
| (3) 3011    | (7) 1781    | (11) 1778   | (15) 1873   | (19) 1771   |
| (4) 1859    | (8) 3007    | (12) 1878   | (16) 1870   | (20) 1855   |

the vicinity of the faulted bus 15. From the speed component of the right eigenvector, the first group is clearly recognizable as that of machines 1826, 1806, and 1859 while the second one represents mainly generators 1854, 1771, and 1855. Note that generator 1826 is the most advanced critical machine picked by SIME in Table IX.

Eigenvalue results for the modes of concern do coincide with Prony analysis in Table V, thus confirming the correctness of the developed algorithms. The fact that the machines' classification listed in this table is not in perfect agreement with that of Fig. 8(b) is quite normal: Table IX lists the machines' classification at the moment the system loses synchronism, **after the fault inception and clearance**, while Fig. 8(b) corresponds to the equilibrium point of the same final system's configuration, **but in the absence of any fault inception**.

In an effort to further verify the appropriateness of the models under varying operating conditions, critical eigenvalues for the modes of concern were computed for various levels of generation in Table VI. In this analysis, the generation level at critical machines singled out by SIME, was adjusted so as to match the generation patterns in Table VI. Eigenvalue results in Table X agree well with previous findings using Prony analysis in Table VI. It should be emphasized that both techniques are

 TABLE X  
EIGENVALUES FOR SEVERAL CMS' GENERATION LEVELS

| $P_C$ (MW) | $\alpha$ (rad/s) | $\omega$ (rad/s) | f (Hz) | $\zeta$ (%) |
|------------|------------------|------------------|--------|-------------|
| 26,103     | -0.2266          | 1.63             | 0.26   | 13.69       |
|            | 0.0354           | 5.61             | 0.89   | -0.63       |
| 25,838     | -0.2359          | 1.66             | 0.26   | 14.01       |
|            | 0.0231           | 5.62             | 0.89   | -0.41       |
| 25,572     | -0.2451          | 1.69             | 0.26   | 14.33       |
|            | 0.0107           | 5.62             | 0.89   | -0.19       |
| 25,306     | -0.2535          | 1.71             | 0.26   | 14.62       |
|            | -0.0018          | 5.62             | 0.89   | 0.03        |

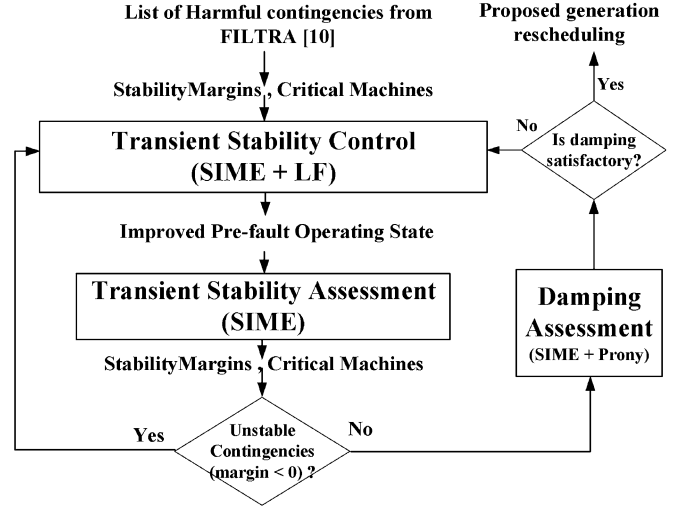


Fig. 9. Integrated SIME-based transient stability and damping assessment and control online function (adapted from [8]).

able to precisely identify the critical generation level at which the system is made stable by shifting generation from critical to noncritical machines. This insight is made possible by the special nature of the developed method. The agreement further helps to establish the confidence in making comparisons between these two fundamentally different analyses.

## VIII. DISCUSSION

### A. On Damping Control Procedures

The damping ratios provided by the control procedure of interarea oscillations of Section VII (see row 4 of Table VI and corresponding curve in Fig. 7) are rather low, as opposed to those of the plant-mode oscillations (see Table III).<sup>6</sup> This is because this procedure is slightly different from the general one (see Section IV-C and its application in Section VI-B). Indeed, it has consistently been using the initial OMIB and corresponding 36 critical machines, while, actually, the group of critical machines changes during the successive iterations, as is shown when extending the maximum integration period from 10 to 15 s (see Table VII).

### B. On Method's Online Capabilities

Fig. 9 displays the damping assessment and control (DA&C) technique proposed in this paper, and shows that it improves the

<sup>6</sup>Even if, in some systems, this damping ratio would be acceptable since they only require it to be positive (see [1]).

online transient stability assessment and control function at the control center (see also Section IV-A).

The resulting online security function would operate cyclically, say, each 30 min. The analysis could be performed in the first 20 min and, as a result, the security function would propose the operator a security-constrained redispatch. The operator would then have the remaining 10 min to decide and apply the proposed generation changes.

DA&C involves only a few burdens additional to that of the SIME-based transient stability assessment and control function, which has been shown to meet the online computing requirements (see [8]–[11]). These additional burdens concern only a small number of time-domain simulations and Prony analysis, which, in turn, raises two main issues. The one relates to the way of specifying main parameters that may change from one system to another or even from one simulation to another.<sup>7</sup> This important issue has already been addressed in [5] and [6].

The other issue concerns the way of filtering the results of Prony from the spurious modes required to fit the signal. This issue may be addressed by preceding Prony with Fourier analysis: Fourier program would indicate the frequency of the actual dominant modes of interest captured by the OMIB, and these would be the only ones that would be taken into account in the results of Prony analysis.

The above solutions make Prony analysis a systematic technique with negligible computing requirements as compared to those of time-domain simulations.

All in all, coupling Prony and SIME methods yields an automatic technique suitable for online applications.

### C. An Interesting Byproduct

The simulation results of Section VII-B suggest that it is possible to detect multiswing instabilities that arise beyond the MIP considered by computing the damping ratio provided by Prony analysis.

For systems experiencing multiswing instabilities, this observation may be exploited advantageously. Indeed, for such systems, the current practice is to use a MIP long enough so as to capture all possible multiswing instabilities. The above observation suggests the possibility of using shorter MIPs by replacing transient stability assessment by damping assessment performed on a shorter MIP. (For example, in Section VII-B, simulations conducted up to 10 s enabled detecting instabilities arising beyond this interval.)

Given that Prony analysis is faster than time-domain simulations, this would allow saving considerable computing times, in particular, for filtering contingencies.

## IX. CONCLUSION

A new approach to online damping assessment and control has been proposed. It relies on Prony analysis and SIME-based transient stability assessment and control.

<sup>7</sup>To simplify the analysis, some parameters like the number of data points to analyze (and, therefore, the sampling rate) can be fixed [18]. Others, like the window length, are set up according to specifics of the power system of concern (e.g., for the EPRI 88-machine system this would be set to the last 5 s of the simulation).

The SIME method combines the functionalities of time-domain programs [accuracy, ability to handle any power system modeling, contingency scenario, and type of instability (plant or interarea mode, first or multiswing oscillations)], together with the functionalities of the equal-area criterion (closed-form expressions of stability margins and stability/instability criteria), and with its own functionalities (identification of the machines relevant to the stability phenomena, along with their degree of relevance).

SIME compresses the postcontingency multimachine system dynamics into the dynamics of a OMIB that passes to Prony for assessing oscillations damping. Further, knowledge of the relevant machines and the interplay between SIME and Prony methods allow improving/stabilizing poorly damped oscillations by suitable generation rescheduling of the precontingency system conditions.

The proposed approach has been investigated, illustrated, and validated by simulations performed on two large systems, which, upon contingency inception, experience plant and interarea mode oscillations.

The investigations have shown that the approach is, indeed, able to improve significantly Prony's technique. Conventional eigenvalue analysis of the postfault equilibrium conditions of the interarea cases confirmed that the proposed method precisely determines the dominant modes of oscillation and the system machines that have a major influence on the ensuing oscillatory phenomena, providing near optimum control solutions for improving/stabilizing poorly damped oscillations.

Admittedly, this control capability is a very needed and valuable complement to transient stability assessment and control.

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